

Topic 14: Redox II

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating redox systems (especially within transition metals), setting up a variety of electrochemical cells, investigating disproportionation in copper(I) salts, performing redox titrations using potassium manganate(VII) or iodine-thiosulfate. Mathematical skills that could be developed in this topic include calculating redox potentials, balancing redox equations from half cells, calculations involving redox titrations.

Within this topic, students can consider how ideas developed in different contexts within chemistry can be shown to be related to a major explanatory principle. Here, cell emfs and equilibrium constants are shown to be related to the fundamental criterion for the feasibility of a chemical reaction: the total entropy change. Students can also consider how chemists continue to search for alternative sources of energy, through the development of fuel cells.

Students should:

1. understand the terms 'oxidation' and 'reduction' in terms of electron transfer, applied to *s*-, *p*- and *d*-block elements
2. understand the terms 'oxidation' and 'reduction' in terms of changes in oxidation number, applied to *s*-, *p*- and *d*-block elements
3. know what is meant by the term 'standard electrode potential', $E^\ominus$
4. know that the standard electrode potential, $E^\ominus$, refers to conditions of:
 - i 298 K temperature
 - ii 100 kPa pressure of gases
 - iii 1.00 mol dm⁻³ concentration of ions
5. know the features of the standard hydrogen electrode and understand why a reference electrode is necessary
6. understand that different methods are used to measure standard electrode potentials of:
 - i metals or non-metals in contact with their ions in aqueous solution
 - ii ions of the same element with different oxidation numbers

CORE PRACTICAL 10: Investigating some electrochemical cells

7. be able to calculate a standard emf, $E^\ominus_{\text{cell}}$, by combining two standard electrode potentials
8. be able to write cell diagrams using the conventional representation of half-cells
9. understand the importance of the conditions when measuring the electrode potential, E
10. be able to predict the thermodynamic feasibility of a reaction using standard electrode potentials

Students should:

11. understand that $E^\ominus_{\text{cell}}$ is directly proportional to the total entropy change and to $\ln K$ for a reaction
12. understand the limitations of predictions made using standard electrode potentials, in terms of kinetic inhibition and departure from standard conditions
13. know that standard electrode potentials can be listed as an electrochemical series
14. understand how disproportionation reactions relate to standard electrode potentials
15. understand the application of electrode potentials to storage cells
16. understand that the energy released on the reaction of a fuel with oxygen is utilised in a fuel cell to generate a voltage
Knowledge that methanol and other hydrogen-rich fuels are used in fuel cells is expected.
17. know the electrode reactions that occur in a hydrogen-oxygen fuel cell
Knowledge of hydrogen-oxygen fuel cells with both acidic and alkaline electrolytes is expected.
18. be able to carry out both structured and non-structured titration calculations including $\text{Fe}^{2+}/\text{MnO}_4^-$, and $\text{I}_2/\text{S}_2\text{O}_3^{2-}$
19. understand the methods used in redox titrations

CORE PRACTICAL 11: Redox titration

Topic 15: Transition Metals

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include the stepwise reduction of vanadium(V) to vanadium(II), investigating the reactions of copper(II) ions or chromium(III) ions, using sodium hydroxide and ammonia solution to identify transition metal ions, investigating autocatalysis, preparing a complex transition metal salt.

Mathematical skills that could be developed in this topic include investigating the geometry of different transition metal complexes.

Within this topic, students can consider the model for the filling of electron orbitals encountered in Topic 1, and see how limitations in that model indicate the need for more sophisticated explanations. They can also appreciate that catalyst research is a frontier area, and one which provides an opportunity to show how the scientific community reports and validates new knowledge.